

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

- **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics, A/B Testing, Hypothesis Testing, Statistical Significance
- **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, matplotlib, seaborn
- **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization
- **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned end-to-end design and development of **cross-platform system integrating** wearable data for 100+ users, enabling scalable **data ingestion** and analytics
- Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%
- Built Python pipeline using pandas and matplotlib to process, visualize longitudinal behavioral datasets, improving analytical accuracy by 43%
- Conducted exploratory analysis on longitudinal behavioral datasets to surface trends and anomalies, presenting findings to stakeholders in structured reports

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Built Python automation scripts using pandas to process and clean high-volume behavioral datasets, reducing analysis turnaround time by 41%
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection
- Documented data sources, metric definitions, and analytical workflows to support team knowledge-sharing and ensure reproducibility across reporting cycles
- Applied hypothesis testing and correlation analysis on social media behavioral data to surface statistically supported insights for stakeholders

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable, microservices-based (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings
- Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval
- Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements
- Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services
- Analyzed large-scale transactional datasets using Python, pandas, and seaborn to identify failure patterns, surface anomalies, and visualize performance trends
- Wrote SQL using window functions, aggregations, and subqueries to map dependencies, detect data quality issues, and speed root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, applying **SQL**, **DAX**, and **statistical analysis** to surface KPI trends and support cross-functional business decision-making
- Automated recurring **Excel**-based reporting workflows using **VBA**, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, improving cross-team visibility and supporting on-time milestone reviews

PROJECTS

FMCG Sales Demand Forecasting

March 2025

View Project

- Analyzed FMCG **sales demand data** across categories and locations to identify trends, seasonal patterns, and key drivers impacting inventory
- Designed A/B tests and applied hypothesis testing to compare ARIMA, SARIMA, and regression models, improving forecast accuracy measured by RMSE and MAPE
- Translated findings into structured stakeholder presentations and written reports on promotions, pricing, and stock behavior to support inventory